

Process Characterization Plan

**A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) —
PCP-007**

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—

Role	Function	Name (synthetic)	Signature	Date
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; Cpk, one-sided process capability index; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LIMS, laboratory information management system; MSAT, Manufacturing Science and Technology; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PD, Process Development; PPQ, process performance qualification; PRESS, prediction error sum of squares; QA, Quality Assurance; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SDM, scale-down model; SEC, size exclusion chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter.

1 Purpose and scope

Cation exchange chromatography is Step 7 of the A-Mab drug substance process and is the polishing step of the purification train. This plan states how the step will be characterized, which parameters will be varied and over what ranges, which attributes will be measured, and by what rules the results will be judged. It is written before the study is executed and reports no findings.

The study will be performed at the sending site on a qualified scale-down model of the commercial column. The commercial process is operated at a production bioreactor scale of 15,000 L, and the purification train is sized to that batch. Every range this study establishes will be derived from data generated on the scale-down model and applied to the commercial process, so the qualification of that model is itself part of the study and is described in §5.1.

The scope covers the 5 operating parameters of the step listed in Table 6.1, the 3 responses measured on the eluate pool, and the 4 quality attributes the step governs, which are listed in Table 4.1. The step forms none of those attributes. Each of them is established upstream, and this step reduces aggregate, host cell protein, residual DNA and leached Protein A towards the levels the drug substance requires. The parameter ranges to be studied are wider than the normal operating ranges of the commercial process, so that the study bounds a knowledge space and not only the routine operating point.

Four activities lie outside this plan. Viral clearance is claimed at the low pH inactivation, anion exchange and virus filtration steps, and no clearance claim will be made for cation exchange, although a cation exchanger does remove virus (CMC Biotech Working Group 2009). Resin life cycle, column packing and sanitization are controlled under SOP-2008 and are qualified in a separate study. Validation of the analytical methods is complete and is recorded in the method validation reports listed in Table 3.2. Confirmation of the ranges at commercial scale belongs to Stage 2 and will be performed during process performance qualification at the receiving site (U.S. Food and Drug Administration 2011).

Prior knowledge from the platform indicates that glycosylation and charge variants are not modified across a cation exchange step operated under platform conditions (CMC Biotech Working Group 2009). Those attributes will therefore not be measured in this study, and the plan makes no claim about them.

2 Objectives

The study has five objectives.

1. To quantify the effect of each studied parameter on the aggregate content, the host cell protein content and the yield of the eluate pool, over the ranges given in Table 6.1.
2. To identify the interactions between those parameters and to fit a predictive model of each response over the region the design covers.
3. To define the region of the multivariate space over which the governed attributes stay within the criteria of §7, and to establish a proven acceptable range for each parameter against each attribute it governs.
4. To classify every parameter of the step under SOP-4001 and to establish whether each normal operating range lies inside the corresponding proven acceptable range.
5. To state the contribution of the step to the control strategy, including the in-process controls and the parameter ranges that will be carried into process performance qualification.

3 Related documents

This plan sits under the transfer plan PTP-001, takes its study scope from the risk assessment RA-001, and instantiates the framework of the master plan PCMP-001. The executed study will be reported in PCR-007, which rolls up into PCMR-001. The package is listed in Table 3.1.

Table 3.1: Related documents in the characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-007	Process Characterization Report (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures and validated methods that govern this study are listed in Table 3.2. SOP-1001 governs the qualification of the scale-down model, SOP-1002 the design and analysis of designed experiments, SOP-2010 the operation of the step, SOP-2008 the resin life cycle and column packing, and SOP-4001 the classification of parameters. The analytical method validation reports are cited by response in §5.3.

Table 3.2: Controlled procedures and validated analytical methods for this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step is operated in bind and elute mode on a strong cation exchange resin. Below its isoelectric point the antibody carries a net positive charge and binds the negatively charged ligand. Species that are acidic at the load pH do not bind and leave in the flow through. The antibody is then eluted by raising the pH of the buffer, which weakens the electrostatic attraction between the antibody and the ligand, and the eluate pool is collected between a start point on the ascending edge of the elution peak and a stop point on its descending edge.

Aggregate is retained more strongly than monomer, because a dimer presents twice the charged surface to the ligand. It therefore elutes after the monomer, and the clearance the step achieves depends on how well the two species are resolved and on where the pool is stopped. Three of the studied parameters act on that resolution. Loading the column more heavily broadens both peaks and brings the aggregate front forward into the monomer. A higher elution buffer pH weakens the binding of both species, brings them off earlier and compresses the separation on which the clearance depends. Collecting further into the descending edge takes the leading part of the aggregate peak into the pool.

Most host cell proteins are acidic at the operating pH and do not bind the cation exchanger, so the bulk of them leaves in the load flow through. The weakly bound remainder is displaced during the wash. A higher wash conductivity shields the charge on the ligand and on the bound species and strips more of that remainder before elution, while the antibody, which binds more strongly, stays bound. A heavier load leaves less free ligand, and host cell protein that the wash would otherwise have displaced co-adsorbs with the antibody. Protein load and wash conductivity act on the same population of weakly bound species, so an interaction between them is expected on the host cell protein content of the pool.

DNA is strongly negatively charged and is repelled by the cation exchange ligand, so it passes through in the load. Its clearance is large and is not expected to depend on the operating parameters. Leached Protein A and its fragments are acidic and do not bind under the load conditions, and the fraction that is complexed with the antibody is released and passed through when the complex binds the ligand. The clearance of leached Protein A is therefore a property of the chemistry of the step and not of any parameter studied here.

Yield is lost at the two edges of the pool, as monomer left in the tail beyond the stop point and as monomer that co-elutes with the aggregate that is cut away. A heavier load broadens the peak and increases both losses. The elution stop collect point is the

direct handle on the trade between aggregate clearance and yield, and quantifying that trade is one of the reasons the step is studied multivariately.

Prior product knowledge for this step is extensive. The same cation exchange resin and the same platform buffer system have been used for three previously licensed humanized IgG1 antibodies, X-Mab, Y-Mab and Z-Mab, and through the A-Mab clinical and engineering campaigns. Across those products the step has behaved as the mechanism above predicts, and its performance has been consistent (CMC Biotech Working Group 2009). Characterization is therefore expected to confirm and bound relationships that are already understood. Prior knowledge is used here to select the parameters, to set the ranges and to state the direction in which each parameter is expected to act, and every quantitative claim in PCR-007 will rest on the data generated under this plan.

The position of the step in the train sets what is at stake. No step downstream of cation exchange reduces aggregate, so the aggregate content of the eluate pool carries through to the drug substance. Host cell protein is cleared again at the anion exchange step, so the criterion this step has to meet for host cell protein is looser than the drug substance specification. Both criteria are derived in §8.

4.2 Quality attributes in scope

Table 4.1: Quality attributes governed by the cation exchange step, with their acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Leached Protein A	process impurity	0-5 ppm	L-M	16

Table 4.1 lists the 4 attributes this step governs. Each is established upstream of the step, and the step clears it. Criticality is assigned by the impact of the attribute on safety and efficacy combined with the uncertainty in that impact, and attributes of high and very high criticality are called critical (CMC Biotech Working Group 2009). Criticality is a continuum, and an attribute of moderate criticality is controlled with the same rigour as a critical one where the process is what sets its level (International Council for Harmonisation 2023).

Aggregate and host cell protein will be carried as responses of the designed experiments. Both are predicted by the mechanism of §4.1 to respond to the parameters of the step, and both are governed by this step to a degree that the criteria of §8 make

demanding. Step yield will be carried as a third response. It is a process performance attribute and not a quality attribute, and it is measured here because it is what the aggregate criterion trades against.

Residual DNA and leached Protein A will not be carried as design responses. The mechanism predicts that neither binds the ligand and that the clearance of both is insensitive to the studied parameters, and the platform history supports that expectation. Both will be measured on the eluate pool of the centre point runs and of the runs at the edges of the design, and the results will be reported in PCR-007 as confirmatory data. Should either show a dependence on a studied parameter, it will be treated as an unexpected result and investigated before the operating region is defined.

4.3 Risk-based prioritization of parameters

RA-001 ranked the parameters of every unit operation before any characterization study was designed, by the method of the case study (CMC Biotech Working Group 2009) and the principles of ICH Q9 (International Council for Harmonisation 2023). Each parameter was scored on its potential impact on a quality attribute and on its potential to interact with other parameters. Where no data or rationale was available to support a lower ranking, the parameter was ranked at the highest level. The scope of this plan is the output of that assessment for the cation exchange step.

The assessment placed the parameters of the step in three groups. 4 parameters were assigned to the multivariate designs, because each acts on a governed attribute and is expected to interact with at least one other parameter. 1 parameter was assigned to univariate assessment, because its effect on the separation is expected to be small across the range the equipment permits. The remaining operating conditions of the step, such as the buffer composition, the bed height and the equilibration volume, are controlled to platform specification and were ranked low. They are verified at buffer release and column release and will not be varied.

The ranges to be studied were set wider than the normal operating ranges of the commercial process. A study that covers only the normal operating range cannot show where performance begins to fail, and a wider range provides the process understanding needed to support movement inside the design space later in the life cycle (International Council for Harmonisation 2009). The ranges in Table 6.1 are the knowledge space of this study, and no claim will be made outside them.

5 Materials and methods

5.1 Scale-down model and its qualification

The study will be run on a laboratory column that models the commercial column. The model will be designed on established principles of chromatography scaling, keeping the bed height, the linear flow velocity, the residence time, and the load, wash and elution volumes normalized to column volumes the same as at commercial scale. Column efficiency will be verified by plate count and peak asymmetry before the column is used, under SOP-2008.

Qualification will follow SOP-1001. Replicate runs at the set-point conditions of Table 6.1 will be performed on the scale-down column, and their results compared with commercial-equivalent data from the engineering and clinical campaigns. The comparison is made on input and output attributes, which are the aggregate and host cell protein content of the load and of the pool, the step yield, and the volume and conductivity of the eluate. The acceptance criteria for the comparison are stated in §7.1. A model that does not meet them will be modified and re-qualified before any characterization run is executed, and the modification will be described in PCR-007.

The centre point runs of both designs are also runs at the set-point conditions. Their replication serves two purposes. It gives the reproducibility of the model under repeat operation, and it provides the pure error term against which lack of fit is tested (§5.5). Because the centre points are executed inside the randomized run order of each design, they also carry any drift in the column, the load or the assays across the execution period.

5.2 Operation

Every run will be executed under SOP-2010. The column will be equilibrated in load buffer, loaded to the protein load of the run, washed at the load and wash conductivity of the run, and eluted with a step change to the elution buffer pH of the run. Pool collection will start on the ascending edge of the elution peak at the platform threshold and will stop on the descending edge at the elution stop collect setting of the run, which is expressed as optical density on the descending edge. The column will then be stripped, cleaned and sanitized under SOP-2008.

The load material will be the neutralized pool of the low pH viral inactivation step, produced from A-Mab harvest at the sending site. A single load lot will be used within a design block wherever material supply permits. The aggregate and host cell protein content of the load will be measured for every run and carried in the data set, because

both set the level from which the step has to clear. Load composition is an input to the step and not a parameter of it, so it is controlled and recorded and is not varied.

Buffers will be prepared to platform specification and released before use. The resin lot, the column dimensions, the packing quality and the number of cycles on the column will be recorded for every run. If the load available for a block differs materially in aggregate or host cell protein content from the material used for model qualification, the difference will be recorded as a deviation at the time it is found and its impact assessed in PCR-007.

5.3 Analytical methods

Each response is measured by one validated method, and all of them are listed in Table 3.2. Aggregate will be measured as high molecular weight species by size exclusion chromatography under AMV-3011. Host cell protein will be measured by enzyme-linked immunosorbent assay under AMV-3012. Residual DNA will be measured by quantitative polymerase chain reaction under AMV-3014, and leached Protein A by enzyme-linked immunosorbent assay under AMV-3016. Protein concentration will be measured by ultraviolet absorbance, and the step yield will be calculated from the product mass in the eluate pool and in the load.

Every sample will be tested by a validated method within its qualified range, and a result outside that range will be reported as such and not extrapolated. The variability of each method is established in its validation report and contributes to the residual variability of the fitted models. The centre point replicates carry process variability and assay variability together, which is the correct basis for the lack of fit test, because the models are fitted to data that carry both.

5.4 Sampling plan

Samples will be taken from the load, the flow through, the wash and the eluate pool of every run. The eluate pool is the sample of record. It will be tested for aggregate, host cell protein and protein concentration in every run of both designs and of the univariate assessment. Residual DNA and leached Protein A will be tested on the eluate pool of the centre point runs and of the runs at the edges of the design, as stated in §4.2. Flow through and wash samples will be retained and will be tested where a pool result needs an explanation.

The screening design includes 3 centre point runs and the response surface design includes 4. Retained samples will be held under the storage conditions qualified in the method validation reports until PCR-007 is approved.

5.5 Statistical methods

The analysis will follow SOP-1002. The factors will be coded so that the set-point is 0 and the edges of the characterization range are -1 and $+1$. Coding puts the estimated effects of parameters measured in different units on a common scale, which is what makes them comparable.

The screening design will be analysed by ordinary least squares on a model containing the main effects and the two factor interactions. An effect will be reported as twice the coded coefficient, which is the change in the response across the full characterization range of the factor. Significance will be judged at $\alpha = 0.05$.

The response surface design will be analysed on the full quadratic model in the same factors, containing the main effects, the two factor interactions and the pure quadratic terms. Model adequacy will be judged on the coefficient of determination (R^2), the adjusted R^2 , the predicted R^2 computed from the prediction error sum of squares, the F test of the regression, and the analysis of variance partition of the residual variance into lack of fit and pure error. The pure error comes from the centre point replicates. The diagnostics reported for each fitted response will be the residuals against the predicted values, a normal quantile plot of the residuals, and the predicted against the actual values.

The screening model identifies which effects are active. The response surface model is the predictive model, and it is the basis of the operating region, of the proven acceptable ranges and of the capability estimate. A two-level design in 4 factors cannot separate curvature from the main effects, so no predictive claim will be made from the screening analysis.

Process capability will be estimated by Monte Carlo simulation of 2,000 batches, with the parameters of the step drawn inside their normal operating ranges and the fitted response surface models used to predict each response. Capability will be reported as a one-sided Cpk against the drug substance criterion of each attribute, because capability states what the process delivers to the specification and not what an intermediate pool has to meet. Every simulation will be run under a fixed random seed, which is recorded in the analysis script, so that a reported value can be reproduced exactly from the same data.

6 Study design

6.1 Factors, ranges and study type

Table 6.1: Parameters of the cation exchange step, their set-points, the range each will be studied over, the normal operating range and the assigned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	25	10-40	18-32	multivariate
Load/Wash conductivity	mS/cm	5	3-7	4-6	multivariate
Elution buffer pH	pH	6	5.8-6.2	5.85-6.15	multivariate
Elution stop collect	OD desc.	1	0.5-1.5	0.7-1.3	multivariate
Elution flow rate	cm/hr	200	100-300	150-250	univariate

Table 6.1 gives the parameters of the step, the set-point of each, the range it will be studied over, its normal operating range and the study type assigned by RA-001. The set-point of every parameter is the centre of its characterization range, and the normal operating range of every parameter lies inside that characterization range. The normal operating range is the range the commercial control system is expected to hold in routine operation, and the characterization range is the range this study will bound.

Table 6.2: Coded factor legend for the multivariate designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

The 4 parameters assigned to the multivariate designs are coded as in Table 6.2. Those codes are used for the design matrices in Appendix A and Appendix B and will be used for the model terms reported in PCR-007.

6.2 Screening design

The screening study will be a two-level full factorial design in the 4 multivariate parameters, augmented with 3 centre point runs. It comprises 16 factorial runs and 19 runs in total. A full factorial estimates all 4 main effects and all 6 two factor interactions without aliasing between them. That matters at this step, because protein load is expected to interact with wash conductivity on host cell protein and with elution buffer pH on aggregate, and a design that confounded either pair with a main effect could not separate the two explanations.

The runs will be performed in randomized order, with the centre point runs distributed through the sequence. The centre points also give a test for curvature. A difference between the mean of the centre points and the mean of the factorial runs indicates that a two-level model does not describe the response, which is the condition the response surface design is there to resolve. The planned matrix is given in Appendix A.

6.3 Response-surface design

The response surface study will be a face centred central composite design in the same 4 parameters. It comprises 16 factorial runs, 8 axial runs and 4 centre point runs, 28 in total. The axial runs are placed on the faces of the cube, so that no run leaves the characterization range of any parameter and no material is produced under conditions this plan has not bounded.

The design adds the pure quadratic terms to the model, and curvature is expected on at least two of the responses. The aggregate content of the pool depends on where the pool is stopped on a peak whose shape changes with load and with elution pH, and a response of that kind is not linear across a range as wide as the one studied here. The yield is bounded above, so it has to curve as the collected fraction is widened. The planned matrix is given in Appendix B.

6.4 Univariate assessment

Elution flow rate will be assessed one parameter at a time. Runs will be performed at the lower edge, the set-point and the upper edge of its characterization range, which are 100, 200 and 300 cm/hr, with the multivariate parameters held at their set-points. The eluate pools will be tested by the methods of §5.3.

The parameter was not placed in the multivariate designs because flow rate acts on resolution through residence time, and at the particle size of the platform resin the change in resolution across the range the equipment permits is expected to be small. The univariate assessment tests that expectation and supports the range. Should it show an effect on aggregate or host cell protein comparable with the effects of the

multivariate parameters, the parameter will be added to a supplementary response surface study before the operating region is defined, and this plan will be revised.

7 Acceptance and decision criteria

This section states the rules by which the study will be judged. They are fixed before execution, and a change to any of them after approval requires a revision of this plan.

7.1 Scale-down model qualification

The scale-down model will be accepted when the pool attributes and the step yield of the qualification runs show no statistically significant difference from the commercial-equivalent data at $\alpha = 0.05$, and when the packed column meets the plate count and asymmetry criteria of SOP-2008. A model that fails either test will be modified and re-qualified, and no characterization run will be executed until it passes. The qualification result will be reported in PCR-007 with the comparison it rests on.

7.2 Model adequacy

A response surface model will be accepted as predictive when the regression is significant at $\alpha = 0.05$, the lack of fit is not significant against the pure error from the centre point replicates, the predicted R^2 is consistent with the adjusted R^2 , and the diagnostics show no structure in the residuals. A model that fails the lack of fit test will be examined for an omitted term or an influential run before any range is derived from it.

Where no factor has a significant effect on a response over the ranges studied, the step is robust for that response across those ranges. No predictive model will be claimed in that case. The ranges will instead be supported by the bound the executed design provides, and the report will say so.

7.3 Response criteria

Table 7.1: Criteria the eluate pool will be judged against in the analysis.

Response	Criterion	Basis
Aggregates (HMW, %)	at or below 1 %	in-process limit, derived in §8.1

Response	Criterion	Basis
Pool HCP (ng/mg)	at or below 193 ng/mg	in-process limit, derived in §8.1
Step yield	no acceptance criterion	process performance, monitored and reported

Table 7.1 gives the criteria the analysis will apply to the eluate pool. The criteria for aggregate and for host cell protein are in-process limits and not the drug substance specifications of Table 4.1, for the reason given in §8.1. Step yield carries no acceptance criterion in this study. It is reported so that the cost of the aggregate criterion is visible, and it will be controlled through the classification of the parameters that govern it.

7.4 Operating region and proven acceptable ranges

The operating region will be declared over the multivariate parameters as the region inside the characterization ranges over which every governed response is predicted to meet its criterion in Table 7.1. Each parameter will additionally carry a proven acceptable range for each attribute it governs, determined as described in §8.

A normal operating range that lies inside the proven acceptable range for every attribute the parameter governs is acceptable as it stands. Where it does not, the normal operating range will be narrowed to lie inside, or the parameter will be classified so that the residual risk is controlled, and PCR-007 will state which was done and why. A parameter whose proven acceptable range is narrower than its characterization range has an edge of failure inside the region studied, and that edge will be reported in the units of the parameter.

7.5 Parameter classification

Every parameter of the step will be classified under SOP-4001 on two questions, whether its variation across the range studied affects a critical quality attribute, and how well it is controlled at commercial scale. A parameter that affects a critical quality attribute and has a low risk of moving outside the design space will be classified as a well-controlled critical process parameter. A parameter that affects a critical quality attribute and carries a higher risk of moving outside the design space will be classified as a critical process parameter. A parameter that affects process performance but no quality attribute will be classified as a key process parameter, and a parameter with no demonstrated effect on either will be classified as a general process parameter (CMC Biotech Working Group 2009).

The classification of every parameter will be stated in PCR-007 with the reason for it, including the parameters for which no significant effect was found. A classification

that rests on the absence of an effect will be reported as resting on that, together with the range over which the absence was demonstrated.

7.6 Completion of the study

The study will be complete when the scale-down model is qualified, both designs and the univariate assessment are executed and analysed, the operating region and the proven acceptable ranges are defined, the capability is estimated and every parameter is classified.

Departures from this plan will be recorded as deviations at the time they occur and dispositioned in PCR-007. A deviation that invalidates a design for the definition of the operating region will require that design to be re-executed on representative material. In that case the invalidated data will be retained and referenced, and the reported analysis will be the re-executed one.

8 Proven acceptable ranges (planned analysis)

8.1 The acceptance basis

The criterion the analysis applies to each attribute is the criterion that applies to the eluate pool of this step, which is not in every case the drug substance specification. The drug substance specification for aggregate is 5 % high molecular weight species and for host cell protein 100 ng/mg (Table 4.1). An intermediate pool has to be judged against a limit that still leaves the downstream steps able to bring the batch to those specifications.

The in-process limit is carried back from the drug substance criterion through the clearance the downstream steps deliver in the nominal train, and is then divided by an assurance margin. The margin is what makes the limit a control rather than a break even point, because a limit set at the undivided ceiling would leave the drug substance dependent on every downstream step delivering its nominal clearance exactly.

No step downstream of cation exchange reduces aggregate. The ceiling for aggregate is therefore the drug substance criterion itself, and after the assurance margin of 5 the in-process limit is 1 % high molecular weight species. Host cell protein is cleared again at the anion exchange step, so its ceiling is above the drug substance criterion, and after the assurance margin of 3.2 the in-process limit is 193 ng/mg. Both limits are given in Table 7.1 and both are fixed before execution.

Residual DNA and leached Protein A will be judged against the drug substance criteria in Table 4.1. They are measured as confirmatory data and are not modelled, so no proven acceptable range will be derived for them.

8.2 The two analyses

A proven acceptable range will be determined for each pair of a governed attribute and a multivariate parameter, in two forms.

The first analysis holds the other parameters at their set-points and evaluates the fitted response surface model for the parameter of interest across its characterization range, on a grid of 201 points. For every parameter in the multivariate designs the set-point is the centre of the characterization range (Table 6.1), so this analysis is performed at the centre of the design in the coded factors. The result is the range

over which the parameter alone, with the rest of the step at target, keeps the attribute within its criterion.

The second analysis propagates normal operating range variation into the prediction. At each of 81 points across the characterization range of the parameter of interest, the other parameters will be drawn 2,000 times from a distribution centred on their set-points whose spread is set by their normal operating ranges and truncated at the characterization range, the fitted model will be evaluated at each draw, and the residual variability of the model will be added. The parameter is acceptable at that point when the 95 % predictive interval of the attribute stays within its criterion. This is a robustness criterion, so the range it returns is the narrower of the two wherever an interaction is active.

8.3 The decision rule and the reporting

The proven acceptable range reported for a parameter is the contiguous interval containing the set-point over which the criterion is met. Where the criterion is met across the whole characterization range, the proven acceptable range is the characterization range, and the report will state that no edge of failure was found inside the region studied. Where the criterion is not met at the set-point itself under an analysis, no range will be reported for that pair and the result will be investigated.

Each governed attribute will be reported with a plot against the parameter that governs it, showing the predicted response across the characterization range with the acceptable region shaded green, and with the set-point and the normal operating range marked. Both analyses will be shown on the same figure, and the full table of ranges will be given for every pair of an attribute and a parameter.

Two bounds apply to a proven acceptable range determined this way. It is a section through a multivariate model taken one parameter at a time, so two ranges may not be assumed to hold together at their edges. The multivariate statement is the operating region of §7.4, and PCR-007 will present it as such. The fitted model also predicts mean levels, so a range derived from the first analysis describes where the mean of the response meets the criterion and carries no assurance about a single batch at its edge. The second analysis is what brings the variation of the process into the range, and the capability estimate of §5.5 is what states the assurance at the operating point.

9 Data management and integrity

Data will be recorded and retained under the ALCOA+ principles. Chromatograms and run records will be captured in the chromatography data system and analytical results in the laboratory information management system. Both systems are validated, both record the analyst and the time of the entry, and the audit trail of each is enabled and reviewed. Raw data will not be transcribed by hand into the analysis.

The analysis will be performed with scripts held under version control, and the data set analysed in PCR-007 will be the data set released by second person review. Every reported value will be traceable to the run record and the assay result it comes from, and every simulation will be run under a fixed random seed, so that the whole analysis can be reproduced from the same data.

Each run record will be reviewed against this plan by a second person before the data enter the analysis. A departure from the plan will be recorded when it is found, with what was done, why it was done and which runs it affects. No result will be excluded from an analysis without a documented technical reason, and any exclusion will be reported in PCR-007 with that reason and with the effect of the exclusion on the model.

10 Roles and responsibilities

Table 10.1: Functions and their responsibilities for this study.

Function	Responsibility
Process Development / MSAT	Owens this plan, qualifies the scale-down model, executes the runs, performs the analysis and writes PCR-007
Analytical Development	Releases the validated methods, performs the testing and reports the assay performance
Statistics	Approves the designs and the analysis method, and reviews the model adequacy judgements
Manufacturing Science, receiving site	Reviews this plan and PCR-007 for fit with the commercial equipment and control system, and receives the ranges for Stage 2
Quality Assurance	Approves this plan and PCR-007, and dispositions deviations

Table 10.1 gives the functions accountable for the study. Process Development owns the plan and the execution, and no function may change a range, a criterion or a design without a revision of this plan approved by the same functions that approved it. The approval record is Table 1.

11 Deliverables and schedule

The deliverable of this plan is PCR-007. It will report the executed study, the operating region, the proven acceptable ranges, the estimated capability, the classification of every parameter and the contribution of the step to the control strategy. PCR-007 rolls up into PCMR-001, which consolidates the campaign for the receiving site.

The study will run in five phases. Qualification of the scale-down model comes first and gates everything after it. The screening design follows, then the response surface design, then the analysis and the report. The univariate assessment may be executed in parallel with the response surface design, because it uses the same load material and the same methods and its result is not needed until the parameter classification is made.

Each phase will be released by a review of the preceding phase against the criteria of §7. No dates are fixed in this plan. The schedule is held in PTP-001 and is coordinated with the transfer of the analytical methods to the receiving site.

12 Risks and assumptions

Four risks to the study are worth stating in advance.

The first is the representativeness of the scale-down model. Every range in PCR-007 will be derived from data generated on a laboratory column, and the claim that those ranges hold at commercial scale rests on the qualification of §5.1. That qualification compares performance at the set-point, and it cannot demonstrate equivalence at the edges of the ranges studied. This is a limitation of the approach and not of its execution, and it is why the ranges are confirmed at scale during Stage 2.

The second is the representativeness of the load. The load is the pool of the preceding step, and its aggregate and host cell protein content vary with upstream performance. A load that is not representative of routine material can produce an effect that is real in the data and wrong for the process. The load will be characterized for every run, and material that is out of trend will be treated as a deviation before the affected data are analysed.

The third is assay variability near a limit. Residual DNA is cleared to a level far below its criterion, and a difference measured there is a difference between two results that both sit near the lower end of the qualified range of the method. Differences of that kind will be attributed to the assay and will not be interpreted as parameter effects.

The fourth is material supply. The study needs load material of consistent quality across 47 design runs and the runs of the univariate assessment. Where more than one load lot is used, the lot will be recorded in the data set and its effect examined before the models are fitted.

Three assumptions underlie the plan. Platform behaviour observed with X-Mab, Y-Mab and Z-Mab is assumed to transfer to A-Mab, which the mechanism of §4.1 supports and which the qualification runs will test at the set-point. The parameters that are not studied are assumed to be controlled to platform specification and to contribute no material variability to the responses. The analytical methods are assumed to be validated across the range of values the study will generate, which the method validation reports of Table 3.2 establish.

13 Approvals

This plan takes effect when it has been approved by the functions listed in Table 1, and no run will be executed before that approval. A change to the design, to the ranges or to the acceptance criteria after approval requires a revision of this plan and a new approval by the same functions. Any departure during execution will be recorded as a deviation and dispositioned in PCR-007.

Appendix A — Planned screening design matrix

The coded settings of the 19 planned screening runs are given in Table 13.1. The factor codes are those of Table 6.2. The coded values -1 and $+1$ are the edges of the characterization range of each factor in Table 6.1, and 0 is its set-point. No response column is shown, because no run has been executed.

Table 13.1: Planned screening design matrix, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Appendix B — Planned response-surface design matrix

The coded settings of the 28 planned response surface runs are given in Table 13.2, in the same coding as Appendix A. The axial runs carry one factor at its face value and the remaining factors at 0.

Table 13.2: Planned response-surface design matrix, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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